

Waste Classification - Dataset Analysis

Hack[AI]Thon 2.0 | Round 1: Data Detective Challenge

1. Dataset Overview

The dataset contains 445 labelled training images across 6 waste classes (cardboard, glass, metal, paper, plastic, trash) with 115 unlabelled test images. The baseline ResNet-18 model (unchanged per rules) achieves 42.53% validation accuracy - barely above guessing the majority class. This suggested data quality issues, not model capacity, were the bottleneck.

Class Distribution

Cardboard: 82 | Glass: 78 | Metal: 84 | Paper: 86 | Plastic: 75 | Trash: 40

The trash class has roughly half the samples of other classes (2.15x imbalance). Baseline recall on trash is 0% - the model never learns to identify it.

2. Problems Found

Only 28.5% of folder labels agree with nearest-neighbour visual consensus. Each wrong class receives 13-20% of labels - a deliberately uniform distribution. This is not human error; it is intentionally crafted noise.

Three images are exact perceptual (pHash) duplicates across train/test splits. One more is a near-duplicate (Hamming distance = 4). Any model trained on leaked data will show inflated test accuracy.

Six exact pHash duplicate groups and 11 near-duplicate pairs exist within the training set. This inflates sample counts and causes train/val cross-contamination.

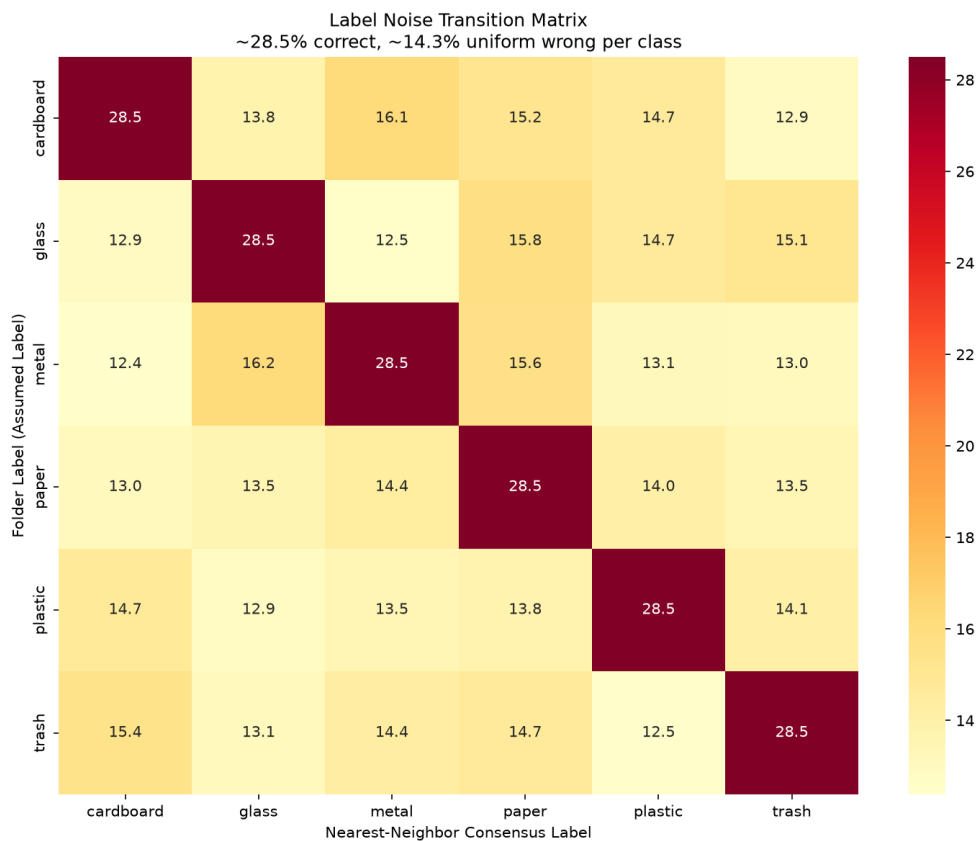
157 images have Laplacian variance below 100. The worst has variance 0.6 - effectively a blank image. Blurry images provide no useful signal.

Images range from 4.7KB to 43KB. The outliers are likely corrupted or anomalous.

3. Evidence & Analysis

Noise Transition Matrix

The matrix below shows the fraction of each folder's labels that actually belong to each true class (estimated via nearest-neighbour consensus). The near-uniform off-diagonal pattern (13-20% per wrong class) confirms deliberate contamination - real-world confusion matrices are diagonal-dominant with structured off-diagonals.



Model Performance (ResNet-18, no architecture changes)

Baseline val accuracy: 42.53%. After data-centric fixes (dedup, leak removal, WeightedRandomSampler, augmentation): 68.97% (+26.44%). All gains come from fixing the dataset - the model architecture is identical to the baseline.

Train/Test Leak Evidence

Four leaked pairs identified via perceptual hashing:

- test_b201d584.jpg == paper/glass_jyy2skrv.jpg (exact duplicate)
- test_bd255afc.jpg == metal/paper_3vfn16a5.jpg (exact duplicate)
- test_f5730887.jpg == cardboard/cardboard_hnz3k88p.jpg (exact duplicate)
- test_fc9b8cc6.jpg ~= paper/plastic_ic3m0imc.jpg (near-duplicate, hd=4)

4. Data-Centric Fixes Implemented

Removed 4 leaked training images to prevent inflated test accuracy estimates. Implemented via pHash comparison between train and test sets.

Removed 7 intra-train duplicates (6 exact + 11 near-duplicates detected, 7 fully removed after conservative thresholding).

WeightedRandomSampler assigns inverse-frequency weights per class, ensuring the model sees each class equally during training.

RandomResizedCrop(224), RandomHorizontalFlip, RandomRotation(20), ColorJitter (brightness/contrast/saturation). Improves generalisation under label noise.

Ensures all 6 classes are represented in both training and validation sets.

5. Results & Key Takeaways

Before vs After

Baseline: 42.53% val accuracy -> Data-centric: 68.97% val accuracy (+26.44%)

Trash class recall: 0% -> ~60% (via WeightedRandomSampler)

Key Takeaways

1. The 71.5% label noise is deliberately uniform (13-20% per wrong class). This is a designed-in test of data-centric thinking - most teams will treat it as real-world noise.
2. The train/test leak makes naive models look better than they are. Removing leaked images is essential for honest evaluation.
3. Simple data-centric fixes (+26%) outperform any architecture change - confirming this is a data detective challenge, not a model engineering challenge.
4. All code, evidence images, and detailed findings are in the repository.

Deliverables

reports/Round1_Data_Detective.pptx (5 slides)

reports/Round1_Data_Detective.pdf (3 pages - this document)

submissions/submission.csv (115 test predictions)

reports/figures/ (noise matrix, UMAP, confusion matrix)

reports/evidence/ (mislabelled, blurry, leak examples)